Class 10 Maths Formula Sheet

Youtube Channel - **Mukesh Agarwal Maths**



Chapter 1 : Real Number

Fundamental Theorem of Arithmetic

Every Composite Number can be expressed as a product of primes and this expression is unique,except the order in which it is written.

Composite Number = Product of Primes

Factor of a Number :

If a Number divides another number exactly, then the number which divides, is called a factor of that number.

Real Number

- **Rational** — Can be Written in p/q Form
  - If $q = 2^m \times 5^n$ → **Terminating** (If Prime Factor of Denominator is Only 2 & 5)
    - $\frac{13}{8} = \frac{13}{2^3} = 1.625$
    - $\frac{17}{4} = 4.25$
    - $\frac{57}{25} = 2.28$
  - If $q \neq 2^m \times 5^n$ → **Non Terminating Repeating** (If Prime Factor of Denominator Contains any other Number other than 2 or 5)
    - $\frac{29}{6} = 4.8333...$
    - $\frac{1763}{1800} = \frac{1763}{2^3 \times 5^2 \times 3^2} = 0.979444.....$
- **Irrational** — Cannot be Written in p/q Form → **Non Terminating Non Repeating**
  - $12.3450748193...$
  - $\sqrt{2} = 1.414213...$
  - $\pi = 3.1415926...$
  - $\sqrt{\text{NonPerFect Square}}$
  - $\sqrt[3]{\text{NonPerFect Cube}}$

HCF and LCM

1. For any Two Positive Integers a & b,
   HCF(a,b) × LCM(a,b) = a×b

   Note : This Formula is Not applicable for More than 2 Numbers

2. LCM of Fraction $= \frac{\text{LCM of Numerator}}{\text{HCF of Denominator}}$

3. HCF of Fraction $= \frac{\text{HCF of Numerator}}{\text{LCM of Denominator}}$

4. If One Number is **factor** of other Number, then **Larger Number is LCM & Smaller Number is HCF** of two Numbers

Formula to Quickly Solve Word Problems

HCF → Same Remainder $HCF(N_1 - R, N_2 - R, N_3 - R)$

HCF → Diff Remainder $HCF(N_1 - R_1, N_2 - R_2, N_3 - R_3)$

LCM → Same Remainder $LCM(N_1, N_2, N_3) + R$

LCM → Diff Remainder $LCM(N_1, N_2, N_3) - (N_1 - R_1)$

Condition Applied $(N_1 - R_1) = (N_2 - R_2) = (N_3 - R_3)$

Watch this Video , To Understand These Formulas
https://youtu.be/A6biZlw8ltA

Operation on Rational & Irrational Number

1. $\left(\text{Rational Number}\right) \overset{+}{\times} \underset{\div}{-} \left(\text{Rational Number}\right) = \left(\text{Rational Number}\right)$
2. $\left(\text{Rational Number}\right) \overset{+}{\times} \underset{\div}{-} \left(\text{Irrational Number}\right) = \left(\text{Irrational Number}\right)$
3. $\left(\text{Irrational Number}\right) \overset{+}{\times} \underset{\div}{-} \left(\text{Irrational Number}\right) = \left(\text{Rational or Irrational}\right)$

Ch 7 : Coordinate Geometry

1. Distance Formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

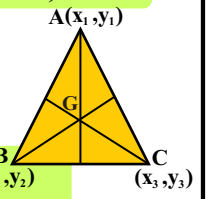
A (x_1, y_1) — B (x_2, y_2)

2. Section Formula $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

A (x_1, y_1) — m — P (x, y) — n — B (x_2, y_2)

3. Mid Point Formula $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

4. Centroid of Triangle $= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$



5. Area of Triangle Formula $= \frac{1}{2}\left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right|$

Condition for Collinearity

6. Let there be Three Points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$

(I). Points A,B & C are collinear if **Area of ΔABC = 0**

(II). Calculate the Distance between AB, BC & AC.
If Sum of Any two distance is equal to third distance, then the 3 points are collinear

AB+BC=AC (or) AC+BC=AB (or) BA+CA=BC

A B C — A C B — B A C

(III). Slope Method - Not to be Studied in Class 10

$\frac{y_1 - y_2}{x_1 - x_2} = \frac{y_1 - y_3}{x_1 - x_3}$

Rectangular Coordinate System

A **coordinate system** is a system of assigning addresses for positions in the plane or in space

The **Plane** is called Cartesian Plane and Both the axis are together called **coordinate axes.**

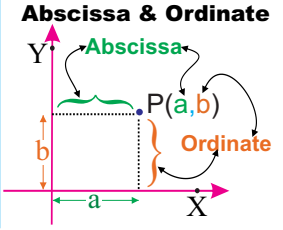
In Coordinate Geometry, **Plane is called Cartesian Plane because** the idea was developed by the Mathematician Rene Descartes(1596 - 1650) who was also known as Cartesius. It is a plane with a rectangular coordinate system that associates each point in the plane with a pair of numbers.

Origin - the point of intersection of the axes

Quadrant - The Coordinate axes divide the plane into 4 parts called quadrants

Absicssa -
Distance of a point from the y-axis (or x-coordinate of a point)

Ordinate -
Distance of a point from the x-axis (or y-coordinate of a point)



Chapter 2 : Polynomial

Polynomial in Simple Terms:

An expression which is composed of **variables, constants and exponents**, that are combined using the mathematical operations such as addition, subtraction, multiplication and division (**Note :** No division by a variable).

Condition for Polynomial :

For the Expression to be called a polynomial ,
i) Power of Variable must not be in **Fraction** &
ii) Power of Variable must not be **Negative**

Degree : Highest Power of Variable

Zeroes : A real number 'k' is called a zero of the Polynomial p(x) , if p(k) =0

Geometrical Meaning of Zeroes:

Zeroes of a Polynomial is the x-coordinates of Points where Graph intersects the x-axis

Number of Zeroes :

i) Number of Zeroes of a polynomial is equal to Number of Times the Graph intersects the x-axis
ii) Number of Zeroes ≤ Degree of Polynomial
iii) A polynomial of degree n has atmost n zeroes.

Classification of Polynomials

On The Basis of Degree	On The Basis of Terms
Degree 0 - Constant Degree 1 - Linear Degree 2 - Quadratic Degree 3 - Cubic Degree 4 - Biquadratic (Quartic)	1 Term - Monomial $5x^2$, 9x, 4 2 Unlike Term – Bionomial $7x+3$, $8x^3+x^2$ 3 Unlike Term - Trinomial $7x + 4 - 5x^2$

Formula of Making Quadratic Polynomial

If α and β are zeroes of Quadratic Polynomial , then

$= k[x^2 - (\text{Sum of Zeroes})x + \text{Product of Zeroes}]$

$= k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right]$

$= k(x-\alpha)(x-\beta)$

Formula of Making Cubic Polynomial

$= x^3 - (\text{Sum of Zeroes})x^2 + (\text{Sum of Product of Zeroes})x - (\text{Product of Zeroes})$

$= x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta+\beta\gamma+\gamma\alpha)x - \alpha\beta\gamma$

$= k(x-\alpha)(x-\beta)(x-\gamma)$

Relation between Zeroes & Coeffecient of Quadratic Polynomial

$$ax^2 + bx + c = 0,\ a \neq 0$$

→ Sum of Zeroes $(\alpha+\beta) = \dfrac{-\text{Coeff of } x}{\text{Coeff of } x^2} = \dfrac{-b}{a}$

→ Product of Zeroes $(\alpha\beta) = \dfrac{\text{Const Term}}{\text{Coeff of } x^2} = \dfrac{c}{a}$

Very Impt Note : Multiplying a Polynomial by a Constant does not affect its Zeroes

Relation B/w Zeroes & Coeff of Cubic Polynomial

$$ax^3 + bx^2 + cx + d = 0,\ a \neq 0$$

→ Sum of Zeroes $(\alpha+\beta+\gamma) = \dfrac{-\text{Coeff of } x^2}{\text{Coeff of } x^3} = \dfrac{-b}{a}$

→ Sum of Prod of Zeroes $(\alpha\beta+\beta\gamma+\gamma\alpha) = \dfrac{\text{Coeff of } x}{\text{Coeff of } x^3} = \dfrac{c}{a}$

→ Product of Zeroes $(\alpha\beta\gamma) = \dfrac{-\text{Const Term}}{\text{Coeff of } x^3} = -\dfrac{d}{a}$

Super Important Fact

Relation b/w (Sign of Coeff) & (Sign of Roots) of Quadratic Polynomial

	Sign of Roots	a	b	c
Case 1	Both Roots Positive $\alpha = +ve, \beta = +ve$	+ −	− +	+ −
Case 2	Both Roots Negative $\alpha = -ve, \beta = -ve$	+ −	+ −	+ −
Case 3	One Positive, Other Negative Mag (+ve Root) > Mag(-ve Root)	+ −	− +	− +
Case 4	One Positive, Other Negative Mag (-ve Root) > Mag(+ve Root)	+ −	+ −	− +

Common Formulas

1. $(a+b)^2 = a^2 + b^2 + 2ab$
2. $(a-b)^2 = a^2 + b^2 - 2ab$
3. $a^2 - b^2 = (a+b)(a-b)$
4. $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
5. $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
6. $(a^3 - b^3) = (a-b)(a^2+b^2+ab)$
7. $(a^3 + b^3) = (a+b)(a^2+b^2-ab)$
8. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
9. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ac)$
10. $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)(a-b)^2(b-c)^2(c-a)^2$
11. $(a+b+c)^3 = a^3 + b^3 + c^3 + 3(b+c)(c+a)(a+b)$

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Important Formulas

12. $(a-b)^2 = (a+b)^2 - 4ab$

$\Rightarrow (a-b) = \sqrt{(a+b)^2 - 4ab}$

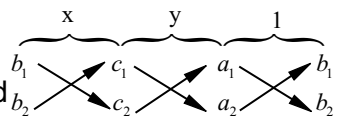
13. $(a+b)^2 = (a-b)^2 + 4ab$
14. $a^2 + b^2 = (a+b)^2 - 2ab$
15. $a^4 + b^4 = [(a+b)^2 - 2ab]^2 - 2a^2b^2$

Ch 3 : Linear Eq in 2 Variable

Standard Form $a_1x + b_1y + c_1 = 0$,
$a_2x + b_2y + c_2 = 0$

Method of Solving Linear Eq in 2 Variable

1. Graphical Method
2. Algebraic Method
   - (i).Substitution Method
   - (ii).Elimination Method
   - (iii).Cross Multiplication Method



$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \quad y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

Condition for Solvability of Linear Eq

Condition	Number of Solution	Graphical Representation	Graph
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Unique Solution	Intersecting Lines	
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Infinite Many Solution	Coincident Lines	
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	No Solution	Parallel Lines	

Homogeneous System of Equations

$a_1x + b_1y = 0$, $a_2x + b_2y = 0$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Unique Solution (or Zero Solution) (i.e x=0,y=0)	Intersecting Lines	
$\frac{a_1}{a_2} = \frac{b_1}{b_2}$	Infinite Number of Non Zero Solutions	Coincident Lines	

Types of Equation & Their Graph

x=a : Graph is Vertical Line parallel to y-axis
y=b : Graph is Horizontal Line parallel to x-axis
x=y : Graph is Straight Line Passing Through Origin at 45°
x= -y : Graph is Straight Line Passing Through Origin at 135°
ax+by+c=0 : Graph is Straight Line

Important

1. In this Equation, ax+by+c=0, if c =0, then Line Passes Through Origin

2. **Upstream =** Boat Speed - Stream Speed — When Boat & Stream are in OPPOSITE Direction
   Downstream = Boat Speed + Stream Speed — When Boat & Stream are in SAME Direction

3. The difference of two number is 4 and the difference of their reciprocal is 4/21. Find the Numbers. **Tricky Question**
   Let Greater Number =x and Smaller Number =y
   $x - y = 4$.....(i)
   Formula Used in this question : $x + y = \sqrt{(x-y)^2 + 4xy}$
   $\frac{1}{y} - \frac{1}{x} = \frac{4}{21}$.....(ii)

Ch 4 - Quadratic Equation

Defination

An equation of the form $ax^2 + bx + c = 0, a \neq 0$ is called a quadratic equation in variable x, where a, b, c are constants

Standard Form : $ax^2 + bx + c = 0, a \neq 0$

Method of Solving Quadratic Equation

(i) Factorization Method (Middle Term Split)
(ii) Quadratic Formula
(iii) Completing the Square Method (Removed)

Quadratic Formula

$$x = \frac{-b \pm \sqrt{D}}{2a}, \text{ where } D = b^2 - 4ac$$

Nature of Roots

Nature of Root Depends on value of Discriminant(D).

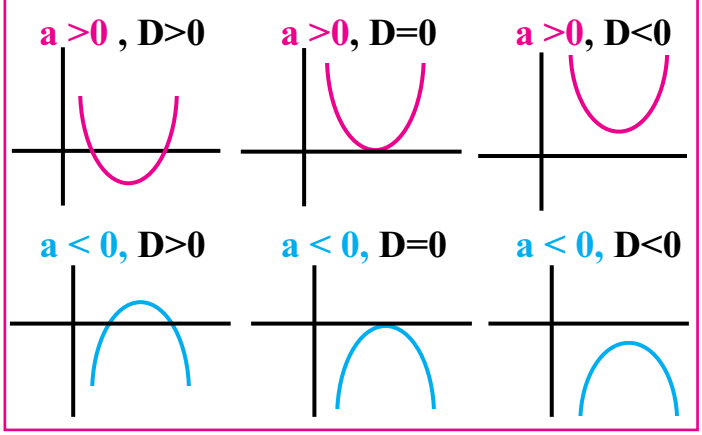
$D = b^2 - 4ac$

D>0	**Roots are Real and Distinct (Unequal)** If D is Perfect Square, The Roots are **Rational** If D is not Perfect Square,Then Roots are **Irrational**
D=0	Real & Equal Roots
D<0	No Real Roots (Imaginary Roots)

Note : Condition For Real Root : $D \geq 0$

Graph of Quadratic Polynomial (called Parabola)

$ax^2 + bx + c = 0, \ a \neq 0$



Super Important Concept

Find the least positive value of k for which $x^2 + kx + 16 = 0$ has real roots.

Quadratic Inequality	If $k^2 < 5^2$ Then $-5 < k < 5$	If $k^2 > 5^2$ Then $k > 5$ or $k < -5$

Secret Facts

In Quadratic Equation : $ax^2 + bx + c = 0$

1. If a+b+c=0, Then Roots = 1,c/a
2. If a+c=b, Then Roots = -1,-c/a
3. If a = c, Then Roots are Reciprocal of Each Other
4. If one root is $(\alpha + \sqrt{\beta})$, then other root must be $(\alpha - \sqrt{\beta})$, provided a,b & c are rational

STOP

Ch 5 - Arithmetic Progression

Arithmetic Progression (AP)

It is a sequence of numbers,in which the difference between any two consecutive numbers is constant.

Difference b/w Progression, Sequence, Series

Both **Progressions & Sequence** are sets of numbers which are arranged in a definite order according to some definite rule.

The **difference between a progression and a sequence** is that a **Progression** has a Specific Formula to calculate its nth term, whereas a **Sequence** can be based on a logical rule like 'a group of prime numbers'

Thus, every progression is a sequence but every sequence is not a Progression.

Series

A series is a sequence of numbers that is added by + signs. The word 'series' is said to represent the sum of the numbers, and not the sum itself

Important Formulas

1. $T_n = a + (n-1)d$
2. $T_n = S_n - S_{n-1}$
3. nth Term From End $= l - (n-1)d$
4. $S_n = \frac{n}{2}(2a + (n-1)d)$
5. $S_n = \frac{n}{2}(a + l)$

Where
a= First Term,
d= Common Difference,
n=Number of Terms
Tn= value of nth Term,
Sn= Sum of n Terms
l=Last Term

Selection of Terms in AP

3 Terms in AP : (a-d),a,(a+d)
4 Terms in AP : (a-3d),(a-d),(a+d),(a+3d)
5 Terms in AP : (a-2d),(a-d),a,(a+d),(a+2d)

Condition for 3 Terms to be in AP

Three quantities a, b, c will be in AP, if the difference b/w two successive terms are equal, i.e., b – a = c – b or 2b = a + c,

Middle Term in AP

If Number of Terms is Odd, Then There is One Middle Term	If Number of Terms is Even, Then There are Two Middle Terms
Middle Term $= \frac{n+1}{2}$	Middle Term $= \left(\frac{n}{2}, \frac{n}{2}+1\right)$

Arithmetic Mean :

If 3 numbers a,b,c are in AP, Then Middle Number 'b' is called Arithmetic Mean of a & c

Important Facts....

1 **nth term of AP is a Linear Expression in n.**
nth term (Tn) = pn+q,
Where "coefficient of n" is **common difference** & (p+q) is **First Term** of the AP
For Example : nth term =7n+3
Then, First Term = (7+3)=10 & c.d=7

2 **Sum of 'n' Term of AP is Quadratic Expression in n**
$S_n = An^2 + Bn$
Where 'twice the coefficient of n^2' is **common difference** & (A+B) is **First Term** of the AP

3 Sum of Terms **Equidistant from Beginning & End** of an AP is always constant &
is equal to the sum of First Term & Last Term (a+*l*)
or in other words
Sum of **nth term from beginning** & **nth term from end** is always constant and is equal to (a+*l*)

4 If the **ratio of Sum of First n Term of two AP's** are given and we have to find the **Ratio of mth Term**, then Replace n by (2m-1) in the original given ratio. **For Example**

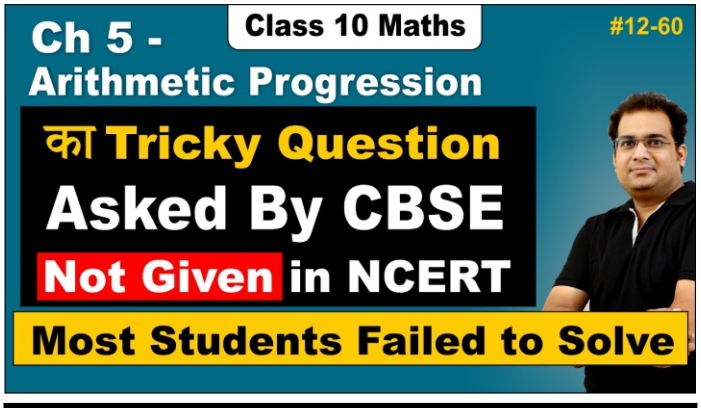
$$\frac{S_n}{s_n} = \frac{7n+1}{4n+27}, \text{ then } \frac{T_m}{t_m} = \frac{7(2m-1)+1}{4(2m-1)+27}$$

Some Standard Results

1. If $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, Then $T_{mn} = 1$
2. If $mT_m = nT_n$, Then $T_{(m+n)} = 0$
3. If $T_m = n$ & $T_n = m$, Then $T_{(m+n)} = 0$
4. If $T_p = q$ and $T_q = p$, Then $T_n = (p+q-n)$
5. If $T_p = a$, $T_q = b$, & $T_r = c$,
   Then $a(q-r) + b(r-p) + c(p-q) = 0$
6. If $S_m = S_n$, Then $S_{(m+n)} = 0$
7. If $S_m = n$ & $S_n = m$, Then $S_{(m+n)} = -(m+n)$
8. If $T_m = \frac{1}{n}$ & $T_n = \frac{1}{m}$, Then $S_{mn} = \frac{(1+mn)}{2}$
9. If $\frac{S_m}{S_n} = \frac{m^2}{n^2}$, Then $\frac{T_m}{T_n} = \frac{2m-1}{2n-1}$

Useful Shortcut (Memorize it)

Sum of First n Natural Number : $1+2+3+.....+n = \frac{n(n+1)}{2}$
Sum of First n Odd Natural Numbers : $1+3+5+.....+(2n-1) = n^2$
Sum of First n Even Natural Numbers : $2+4+6+.....+2n = n(n+1)$

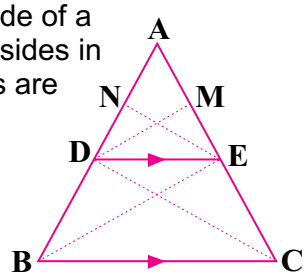


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Chapter 6 - Triangles

1. Thales Theorem
(or BPT- Basic Proportionality Theorem)

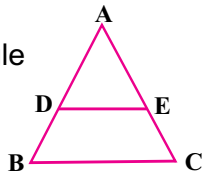
If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.



In Short,
If DE || BC, Then $\frac{AD}{BD} = \frac{AE}{CE}$

2. Converse of BPT Theorem

If a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

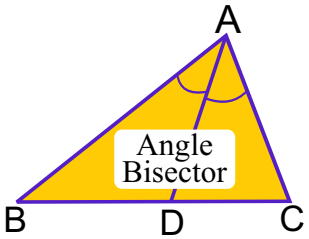


In Short, If $\frac{AD}{BD} = \frac{AE}{CE}$, **Then DE || BC**

3. Angle Bisector Theorem

Internal Bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle.

In Short,
If AD is bisector of ∠A,
Then $\frac{AB}{AC} = \frac{BD}{DC}$

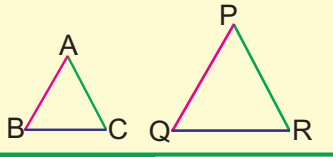


4. Ratio of Areas of Similar Triangles

Theorem 1. The Ratio of areas of two similar triangles is equal to the ratio of the squares of thier **corresponding sides.**

If ΔABC~ΔPQR, Then

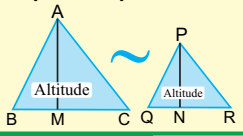
$$\frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{CA^2}{RP^2}$$



Theorem 2. The ratio of the areas of two similar triangles is equal to the ratio of the squares of their **corresponding Altitudes.**

If ΔABC~ΔPQR, Then

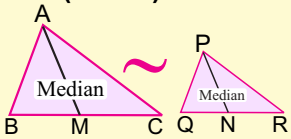
$$\frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{AM^2}{PN^2}$$



Theorem 3. The ratio of areas of two similar triangles is equal to the ratio of the squares of their **corresponding Medians**

If ΔABC~ΔPQR, Then

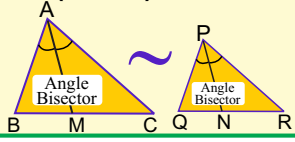
$$\frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{AM^2}{PN^2}$$



Theorem 4. The ratio of areas of two similar triangles is equal to the ratio of the squares of their **corresponding Angle Bisector**

If ΔABC~ΔPQR, Then

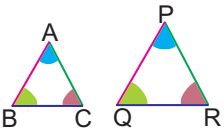
$$\frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{AM^2}{PN^2}$$



Similar Triangles

If two triangles are similar then their corresponding angles are equal & corresponding sides are proportional

i.e If ΔABC~ΔPQR, Then $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$ & $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$

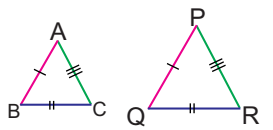


Note : Simiar Triangles have Same Shape, But Not Necessarily the Same Size

Criteria for Similarity of Two Triangles

SSS Similarity

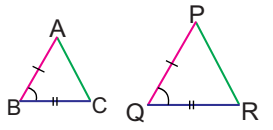
If the three sides of one triangle are respectively proportional to the corresponding three sides of another triangle, then they are similar.



$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$$

SAS Similarity

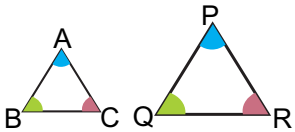
If two sides of a triangle are respectively proportional to the corresponding sides of another triangle and included angle are equal then the triangles are similar.



$\frac{AB}{PQ} = \frac{BC}{QR}$ & $\angle B = \angle Q$

AAA Similarity or (AA)

If the three angles of one triangle are respectively equal to the corresponding angles of another triangle, then the two triangles are similar.



$\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$

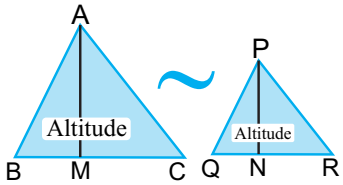
Properties of Similar Triangles

If two triangles are similar, then ratio of **corresponding sides** is equal to the

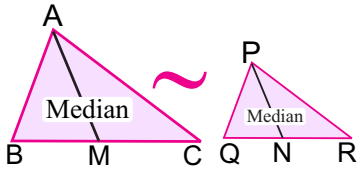
- ratio of corresponding altitudes or
- ratio of corresponding medians or
- ratio of corresponding angle Bisector.



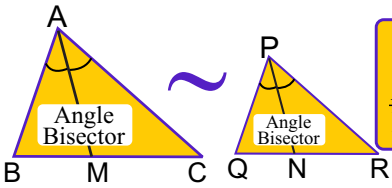
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If ΔABC~ΔPQR, Then $\frac{AB}{PQ} = \frac{\text{Altitude AM}}{\text{Altitude PN}}$



If ΔABC~ΔPQR, Then $\frac{AB}{PQ} = \frac{\text{Median AM}}{\text{Median PN}}$

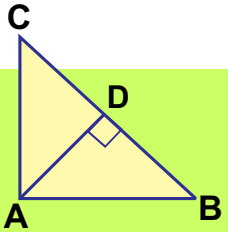


If ΔABC~ΔPQR, Then $\frac{AB}{PQ} = \frac{\text{Angle Bisector AM}}{\text{Angle Bisector PN}}$

An Important Results

In a ΔABC, in Which ∠A = 90, & AD⊥ BC,

Then



(i). $AD^2 = BD.CD$

(ii). $(AB^2+CD^2) = (AC^2+BD^2)$

Ch 8 : Trigonometry

Trigonometry is the Branch of Mathematics in which we study the relationship between the sides & angles of a triangle.

One Trigonometric Ratio in terms of Other T-Ratio

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\cot\theta$	$\sec\theta$	$\operatorname{cosec}\theta$
$\sin\theta$	$\sin\theta$	$\sqrt{1-\cos^2\theta}$	$\frac{\tan\theta}{\sqrt{1+\tan^2\theta}}$	$\frac{1}{\sqrt{1+\cot^2\theta}}$	$\frac{\sqrt{\sec^2\theta-1}}{\sec\theta}$	$\frac{1}{\operatorname{cosec}\theta}$
$\cos\theta$	$\sqrt{1-\sin^2\theta}$	$\cos\theta$	$\frac{1}{\sqrt{1+\tan^2\theta}}$	$\frac{\cot\theta}{\sqrt{1+\cot^2\theta}}$	$\frac{1}{\sec\theta}$	$\frac{\sqrt{\operatorname{cosec}^2\theta-1}}{\operatorname{cosec}\theta}$
$\tan\theta$	$\frac{\sin\theta}{\sqrt{1-\sin^2\theta}}$	$\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}$	$\tan\theta$	$\frac{1}{\cot\theta}$	$\sqrt{\sec^2\theta-1}$	$\frac{1}{\sqrt{\operatorname{cosec}^2\theta-1}}$
$\cot\theta$	$\frac{\sqrt{(1-\sin^2\theta)}}{\sin\theta}$	$\frac{\cos\theta}{\sqrt{(1-\cos^2\theta)}}$	$\frac{1}{\tan\theta}$	$\cot\theta$	$\frac{1}{\sqrt{(\sec^2\theta-1)}}$	$\sqrt{(\operatorname{cosec}^2\theta-1)}$
$\sec\theta$	$\frac{1}{\sqrt{(1-\sin^2\theta)}}$	$\frac{1}{\cos\theta}$	$\sqrt{(1+\tan^2\theta)}$	$\frac{\sqrt{(1+\cot^2\theta)}}{\cot\theta}$	$\sec\theta$	$\frac{\operatorname{cosec}\theta}{\sqrt{(\operatorname{cosec}^2\theta-1)}}$
$\operatorname{cosec}\theta$	$\frac{1}{\sin\theta}$	$\frac{1}{\sqrt{(1-\cos^2\theta)}}$	$\frac{\sqrt{(1+\tan^2\theta)}}{\tan\theta}$	$\sqrt{(1+\cot^2\theta)}$	$\frac{\sec\theta}{\sqrt{(\sec^2\theta-1)}}$	$\operatorname{cosec}\theta$

Values of T Ratios

	0°	30°	45°	60°	90°
Sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ & $\cot\theta = \frac{\cos\theta}{\sin\theta}$

NOTE : $\sin^2\theta \neq \sin\theta^2$

But $\sin^2\theta$ is same as $(\sin\theta)^2$

$\tan1^\circ \tan2^\circ \tan3^\circ \ldots \tan89^\circ = 1$

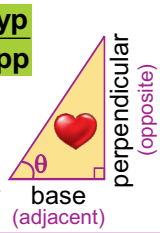
$\cos1^\circ \cos2^\circ \cos3^\circ \ldots \cos90^\circ = 0$

Trigonometry Ratios

(i) $\sin\theta = \frac{p}{h} = \frac{\text{opp}}{\text{hyp}}$, (iv) $\operatorname{cosec}\theta = \frac{h}{p} = \frac{\text{hyp}}{\text{opp}}$

(ii) $\cos\theta = \frac{b}{h} = \frac{\text{adj}}{\text{hyp}}$, (v) $\sec\theta = \frac{h}{b} = \frac{\text{hyp}}{\text{adj}}$

(iii) $\tan\theta = \frac{p}{b} = \frac{\text{opp}}{\text{adj}}$, (vi) $\cot\theta = \frac{b}{p} = \frac{\text{adj}}{\text{opp}}$



Reciprocal Relation

(i) $\sin\theta = \frac{1}{\operatorname{cosec}\theta}$, $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$

(ii) $\cos\theta = \frac{1}{\sec\theta}$, $\sec\theta = \frac{1}{\cos\theta}$

(iii) $\tan\theta = \frac{1}{\cot\theta}$, $\cot\theta = \frac{1}{\tan\theta}$

Complementary Angle

$\sin(90-\theta) = \cos\theta$

$\cos(90-\theta) = \sin\theta$

$\tan(90-\theta) = \cot\theta$

$\cot(90-\theta) = \tan\theta$

$\operatorname{cosec}(90-\theta) = \sec\theta$

$\sec(90-\theta) = \operatorname{cosec}\theta$



$-1 \le \sin\theta \le 1$

$-1 \le \cos\theta \le 1$

Trigonometry Identities

$\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta = 1 - \cos^2\theta$

$\cos^2\theta = 1 - \sin^2\theta$

$\operatorname{cosec}^2\theta - 1 = \cot^2\theta$

$\operatorname{cosec}^2\theta = 1 + \cot^2\theta$

$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\operatorname{cosec}\theta - \cot\theta = \frac{1}{\operatorname{cosec}\theta + \cot\theta}$

$\operatorname{cosec}\theta + \cot\theta = \frac{1}{\operatorname{cosec}\theta - \cot\theta}$

$\sec^2\theta - 1 = \tan^2\theta$

$\sec^2\theta = 1 + \tan^2\theta$

$\sec^2\theta - \tan^2\theta = 1$

$\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$

$\sec\theta + \tan\theta = \frac{1}{\sec\theta - \tan\theta}$

Useful Shortcuts

1. $\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$
2. $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$
3. $\sin^8\theta + \cos^8\theta = 1 + 2\sin^4\theta\cos^4\theta - 4\sin^2\theta\cos^2\theta$
4. If $\sin\theta + \operatorname{cosec}\theta = 2$, then $\sin^n\theta + \operatorname{cosec}^n\theta = 2$
5. If $\cos\theta + \sec\theta = 2$, then $\cos^n\theta + \sec^n\theta = 2$
6. If $\tan\theta + \cot\theta = 2$, then $\tan^n\theta + \cot^n\theta = 2$
7. If $a\sin\theta + b\cos\theta = c$, Then $a\cos\theta - b\sin\theta = \pm\sqrt{a^2+b^2-c^2}$
8. If $a\sin\theta - b\cos\theta = c$, Then $a\cos\theta + b\sin\theta = \pm\sqrt{a^2+b^2-c^2}$
9. If $a\cos\theta + b\sin\theta = c$, Then $a\sin\theta - b\cos\theta = \pm\sqrt{a^2+b^2-c^2}$
10. If $a\cos\theta - b\sin\theta = c$, Then $a\sin\theta + b\cos\theta = \pm\sqrt{a^2+b^2-c^2}$
11. If $\sin\theta + \cos\theta = c$, Then $\sin\theta - \cos\theta = \pm\sqrt{2-c^2}$
12. If $\sec\theta + \tan\theta = p$, Then $\sec\theta = \frac{1}{2}\left(p + \frac{1}{p}\right)$
13. If $\sec\theta - \tan\theta = p$, Then also $\sec\theta = \frac{1}{2}\left(p + \frac{1}{p}\right)$
14. If $\operatorname{cosec}\theta + \cot\theta = p$, Then $\operatorname{cosec}\theta = \frac{1}{2}\left(p + \frac{1}{p}\right)$
15. If $\operatorname{cosec}\theta - \cot\theta = p$, Then also $\operatorname{cosec}\theta = \frac{1}{2}\left(p + \frac{1}{p}\right)$

Ch 9 : Height & Distance

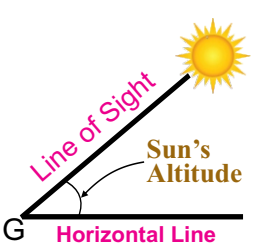
Horizontal Line : If one looks up at some object without raising up or lowering down his head, the line of sight is called horizontal line.

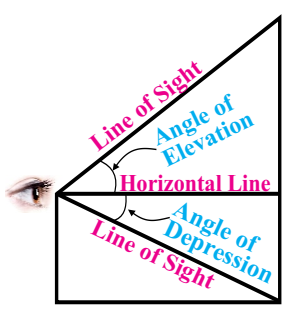
Line of Sight : It is a straight line Between Persons's Eye and the Point(either above or below or anywhere) at which a person is looking.

Angle of Elevation : If we raise our head(or Eye) to see an object then angle between line of sight and horizontal line is called Angle of Elevation.

Angle of Depression : If we lower down our head(or Eye) to see an object then angle between line of sight and horizontal line is called Angle of Depression

Sun's Altitude : If the sun is seen from a point G then the angle between HORIZONTAL LINE & Line Of Sight is called Sun's Altitude. Thus Angle of Elevation of the Sun from a point P is called Sun's Altitude.







Youtube Channel

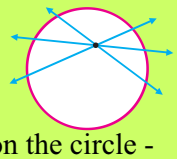
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Ch 10 : Circles

Circle is collection of all the points which are equidistant from a fixed Point called Centre.

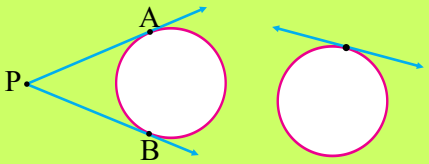
Tangent : It is a line which intersect the circle at only one point. The Point at which the tangent line intersects the circle is called the **point of contact.**

Number of tangents to a Circle :



(i) When the point lies inside the circle
- No Tangent Possible

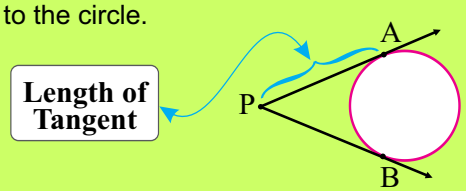
(ii) When the point lies on the circle - There is one & only one tangent passing through that Point



(iii) When point lies outside the circle - There are exactly two tangents Possible.

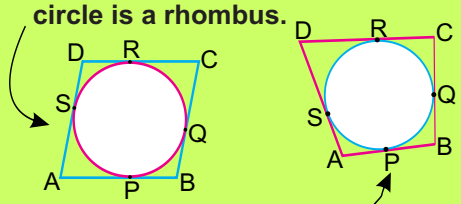
Length of a Tangent :

the length of the segment of the tangent from the external point P to the point of contact with the circle is called the length of the tangent from the point P to the circle.



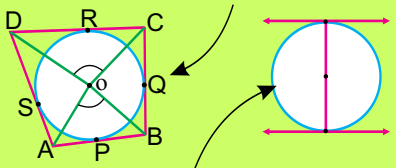
Some Standard Results of Circle

(I) Parallelogram circumscribing a circle is a rhombus.



(ii) If a quadrilateral ABCD is drawn to circumscribe a circle.
Then **AB + CD = AD + BC**

(iii) Opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at centre of circle. i.e ∠AOB + ∠COD=180°

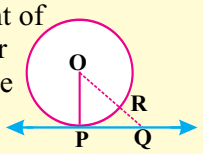


(iv) Tangents drawn at the ends of a diameter of a circle are parallel.

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Theorem 1.

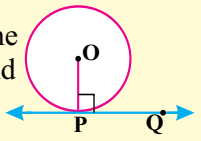
The tangent at any point of a circle is perpendicular to the radius through the point of contact



i.e If PQ is Tangent & OP is Radius, then ∠OPQ=90°

Theorem 2.

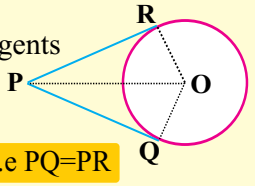
A line drawn through the end point of a radius and perpendicular to it, is a tangent to the circle



i.e If OP is Radius & ∠OPQ=90° then PQ is Tangent

Theorem 3.

The lengths of tangents drawn from an external point to a circle are equal.

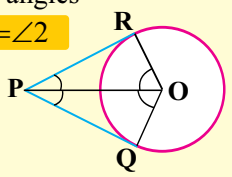


i.e PQ=PR

Theorem 4.

If two tangents are drawn to a circle from an external point then

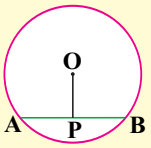
(I) they subtend equal angles at the centre. i.e ∠1=∠2



(ii) they are equally inclined to the line segments joining the centre to that point. i.e ∠3=∠4

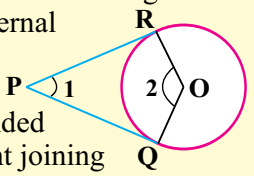
Theorem 5.

In two concentric circles,the chord of the larger circle,which touches the smaller circle, is bisected at the point of contact. i.e AP=PB



Theorem 6.

The angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.

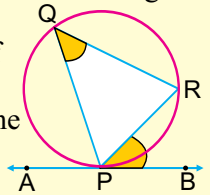


i.e ∠1+∠2=180°

Alternate Segment Theorem



The angle formed between the tangent and the chord through the point of contact of the tangent is equal to the angle formed by the chord in the alternate segment".

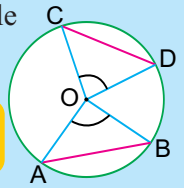


The alternate segment theorem is also known as the **tangent-chord theorem.**

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Some Theorems of Circles of Class 9

1. Equal chords of a circle subtend equal angles at the centre.

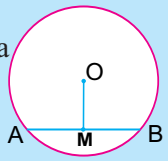


i.e if AB=CD, Then ∠AOB =∠COD

2. If the angles subtended by the chords of a circle at the centre are equal, then the chords are equal.

i.e if ∠AOB =∠COD, Then AB = CD

3. The perpendicular from the centre of a circle to a chord bisects the chord.

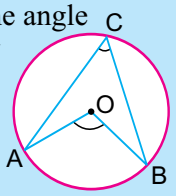


i.e if OM ⊥ AB, Then, AM = BM

4. The line joining the centre of a circle & mid point of chord is perpendicular to the chord.

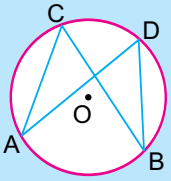
if AM=BM, Then OM ⊥ AB

5. The angle subtended by an arc at the centre is double the angle subtended by it at any point on remaining part of the circle.



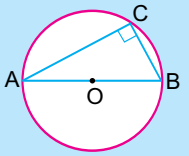
∠AOB =2∠COD

6. Angles in the same segment of a circle are equal.



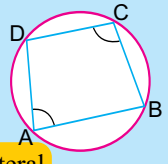
i.e ∠ACB =∠ADB

7. Angle in a semicircle is a right angle.



i.e ∠ACB = 90°

8. The sum of either pair of opposite angles of a cyclic quadrilateral is 180°

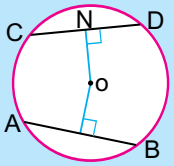


If ABCD is cylic Quadrilateral Then, ∠A+∠B = 180°

9. If the sum of a pair of opposite angles of a quadrilateral is 180°, then the quadrilateral is cyclic.

If, ∠A+∠B = 180° Then ABCD is cylic Quadrilateral

10. Equal chords of a circle are equidistant from the centre.

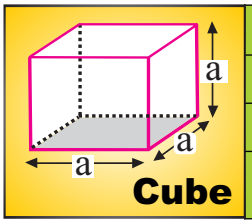


i.e If AB=CD, Then OM=ON

11. Chords equidistant from the centre of a circle are equal in length.

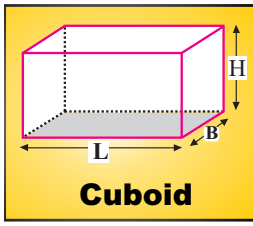
i.e If OM=ON, Then AB=CD

Ch 12 – Surface Area & Volume



Cube

LSA	$4(\text{Side})^2$
TSA	$6(\text{Side})^2$
Volume	$(\text{Side})^3$
Diagonal of Cube	$\text{side}\sqrt{3}$

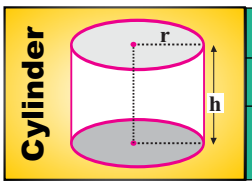


Cuboid

LSA	$2(L+B)\times H$
TSA	$2(LB+BH+LH)$
Volume	$L\times B\times H$
Diagonal of Cuboid	$\sqrt{L^2+B^2+H^2}$

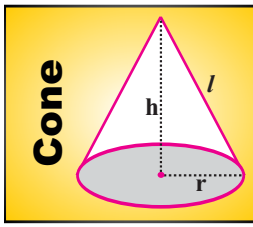
kilo
hecto
deca
meter
deci
centi
mili

Common Pythagorean Triplets
3,4,5
5,12,13
8,15,17
9,40,41
20,21,29



Cylinder

CSA	$2\pi rh$
TSA	$2\pi rh+2\pi r^2$
Volume	$\pi r^2 h$

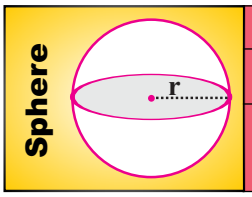


Cone

CSA	πrl
TSA	$\pi rl+\pi r^2$
Volume	$\frac{1}{3}\pi r^2 h$
Slant Height	$l=\sqrt{r^2+h^2}$

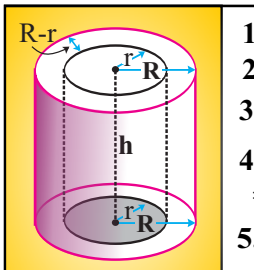
Volume Conversion
$1m^3$ = 1000 Litre
$1cm^3$ = 1/1000 Litre
$1000cm^3$ = 1 Litre
1 Hectare = $10000\ m^2$

Note :
Litre is not Equal to kg
Litre is Unit of Volume & Kg is Unit of Mass



Sphere

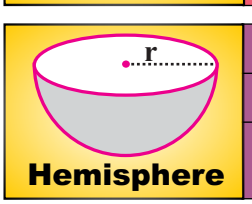
CSA	.....
TSA	$4\pi r^2$
Volume	$\frac{4}{3}\pi r^3$



Hollow Cylinder

1. **Outer CSA** $= 2\pi rh$
2. **Inner CSA** $= 2\pi Rh$
3. **Area of 2 Ring** $=2\pi(R^2-r^2)$
4. **Total Surface Area (TSA)** $=2\pi rh+2\pi Rh+2\pi(R^2-r^2)$
5. **Volume of Solid Portion** $=\pi(R^2-r^2)\times h$
6. **Thickness** $=(R-r)$

To Convert m/sec into km/hr
Multiply By **18/5**
For Example: 10m/sec
$=10\times\frac{18}{5}=36\text{km/hr}$



Hemisphere

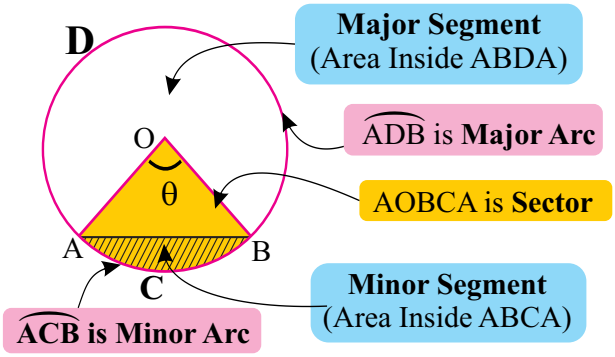
CSA	$2\pi r^2$
TSA	$3\pi r^2$
Volume	$\frac{2}{3}\pi r^3$

Mass or Weight = Volume x Density
Volume of Prism = Area of Base x Height

For More Study Material of Class 10, Check My Website
www.MukeshAgarwalMaths.Com

Ch 11 – Area Related to Circle

Square Root Short Trick
$\sqrt{k}=\sqrt{x\pm y}=\sqrt{x}\pm\frac{y}{2\sqrt{x}}$



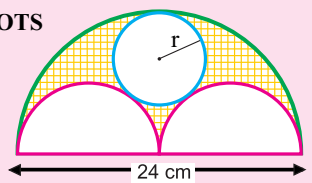
Major Segment (Area Inside ABDA)
$\overset{\frown}{ADB}$ is **Major Arc**
AOBCA is **Sector**
Minor Segment (Area Inside ABCA)
$\overset{\frown}{ACB}$ is **Minor Arc**

1. Arc Length $=\frac{\theta}{360}\times 2\pi r$
2. Area of Sector of Circle $=\frac{\theta}{360}\times\pi r^2$
   $=\frac{1}{2}\times\text{Radius}\times\text{Arc Length}$ **(Impt)**
3. Area of Minor Segment
   $=$ Area of Sector $-$ Area of Triangle
   $=\frac{\theta}{360}\pi r^2-\frac{1}{2}r^2\sin\theta$
4. Area of Major Segment
   $=$ Area of Circle $-$ Area of Minor Segment
5. Area of Triangle $=\frac{1}{2}r^2\sin\theta$
6. Number of Rotation Per Minute
   $=\frac{\text{Distance Covered by Wheel in 1 Minute}}{\text{Circumference of Wheel}}$

Common Formulas

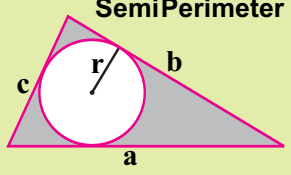
7. Area of Circle $=\pi r^2$
8. Circumference of Circle $=2\pi r$
9. Area of Semi Circle $=\frac{\pi r^2}{2}$
10. Perimeter of Semicircle $=\pi r+2r$
11. Area of Quadrant $=\frac{\pi r^2}{4}$
12. Perimeter of Quadrant $=\frac{2\pi r}{4}+2r$
13. Area of Square $=(\text{Side})^2=\frac{1}{2}(\text{Diagonal})^2$
14. Area of Rectangle $=L\times B$
15. Diagonal of Square $=\text{Side}\sqrt{2}$
16. Diagonal of Rectangle $=\sqrt{L^2+B^2}$
17. Area of Rhombus $=\frac{1}{2}\times d_1\times d_2$ (or) Base $\times$ Height
18. Area of Parallelogram $=$ Base $\times$ Height
19. Area of Equilateral Triangle $=\frac{\sqrt{3}}{4}(\text{side})^2$
20. Height of Equilateral Triangle $=\frac{\sqrt{3}}{2}\times\text{Side}$
21. Area of Triangle $=\frac{1}{2}\times\text{Base}\times\text{Height}=\sqrt{s(s-a)(s-b)(s-c)}$
22. Area of Trapezium $=\frac{1}{2}(\text{Sum of Parallel Sides})\times\text{Height}$
23. Area of **4** Wall $=2(L+B)\times H$

HOTS



Radius (r)
=1/6 Diameter of Outer Semicircle

Inradius $=\frac{\text{Area of Triangle}}{\text{Semi Perimeter}}$



$\sqrt{2}=1.414$
$\sqrt{3}=1.732$
$\sqrt{5}=2.236$

Ch 13 – Statistics

Mean : The Average of All Numbers

Median : is the middle most value in a given data set after it is arranged in ascending order.

Mode : The Number that appears the Most in Given Data

Mean

M-1 Direct Method, $\bar{X} = \frac{\sum fixi}{\sum fi}$

M-2 Assumed Mean Method, $\bar{X} = A + \frac{\sum fidi}{\sum fi}$

M-3 Step Deviation Method, $\bar{X} = A + \left(\frac{\sum fiui}{\sum fi}\right) \times h$

where,

A = Assumed Mean, x_i = Class Mark

$d_i = x_i - A$, $u_i = \left(\frac{x_i - A}{h}\right)$

h = Class Size = Upper Limit − Lower Limit

$\text{Class Mark} = \left(\frac{\text{Lower Limit} + \text{Upper Limit}}{2}\right)$

Median

$$= L + \left(\frac{\frac{n}{2} - pcf}{f}\right) \times h$$

Mode

$$= L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

where,

L = lower limit of Modal/Median Class
n = Sum of Frequencies
h = width of class interval (or Class Size)
f = Frequency of Median Class
pcf = Cumulative Frequency of Class Preceding Median Class
f1 = Frequency of Modal Class
f0 = Frequency of class Preceding to Modal Class
f2 = Frequency of class suceeding to Modal Class

Youtube Channel Mukesh Agarwal MATHS

Median of Ungrouped Data

If n is Odd → $\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ term}$

If n is Even → $\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \text{ term} + \left(\frac{n}{2}+1\right)^{th} \text{ term}}{2}$

Steps to Calculate Median of Grouped Data

Step 1 : Calculate the cumulative frequency of each class.
Step 2 : Calculate n/2 (n is sum of all Frequencies)
Step 3 : Then find the class in which the value of cf is **just greater than n/2**. This is the Median Class.
Step 4 : After Identifying Median Class, Obtain Values of L, h, f, pcf
Step 5 : Put All Values in Formula & Calculate

Note : Emperical Relation bw Mean Median & Mode

Mode = 3Median - 2Mean

Ch 14 – Probability

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Total Number of Possible Outcomes}} = \frac{n(E)}{n(S)}$$

- **Probability :** how likely something is to happen. Probability is a concept which numerically measures the degree of certainity of the occurrence of Events
- **Experiment :** An operation which can produce some well defined outcomes is called an experiment.
- **Random Experiment :** An experiment in which all possible outcomes are known, and the exact outcome cannot be predicted in advance, is called a random experiment.
- **Deterministic Experiment :** An experiment whose outcome may be predicted with certainty beforehand
- **Sample Space :** Set of all possible outcomes of a random experiment.
- **Event :** The collection of all or some of the possible outcomes is called an event
- **Elementary Event :** If a random experiment is performed, then each of its outcomes is known as an elementary event. **The Sum of Probabilities of all the elementary events of an experiment is 1**
- **Equally Likely Events :** Those events which have an equal probability of occuring. For example: Each numeral on a die is equally likely to occur when the die is tossed.
- **Complementary Event :** **Two events are said to be complementary when one event will occur if and only if the other does not. The event (not E) is called the complementary event of E. We Denote (not E) by E′ or $\bar{E}$**
The Sum of probabilities of two complimentary events is 1.

Must Check Short Trick of 3 Coins

https://youtu.be/idoDBhUCu2o

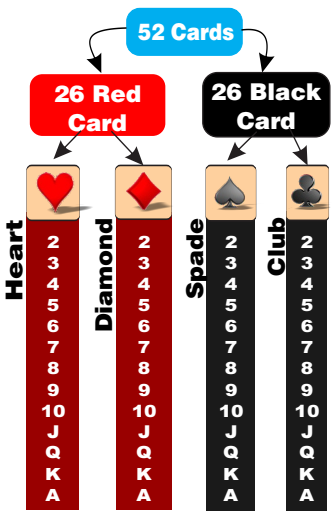
Sample Space when Two Dice are Thrown

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

2 Coins
HH, HT
TH, TT

3 Coins
H H H
H H T
H T H
H T T
T H H
T H T
T T H
T T T

P(Sure Event) =1
P(Impossible Event) =0
$0 \le P(E) \le 1$
$P(\bar{E}) = 1 - P(E)$
$P(E) + P(\bar{E}) = 1$
$P\left(\frac{53 \text{ Sunday}}{\text{in Normal Year}}\right) = \frac{1}{7}$
$P\left(\frac{53 \text{ Sunday}}{\text{in Leap Year}}\right) = \frac{2}{7}$





When 3 Coins are Tossed

P (One Head)	=3/8
P (Two Head)	=3/8
P (Three Head)	=1/8
P (at Least One Head)	=7/8
P (at Least Two Head)	=4/8
P (at Least Three Head)	=1/8
P (at most One Head)	=4/8
P (at most Two Head)	=7/8
P (at most Three Head)	=8/8